

Iris color and macular pigment optical density.

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The present study was designed to assess the relationship between iris color and macular pigment optical density. Both melanin and carotenoids (responsible for iris color and macular pigment composition, respectively) appear to protect the retina through similar mechanisms and higher concentrations may reduce the incidence of retinal degenerations. To evaluate this relationship, 95 subjects were examined and the following variables were measured: iris color; macular pigment optical density (MP); plasma concentrations of lutein and zeaxanthin and beta-carotene; dietary intake of lutein and zeaxanthin and beta-carotene; and total fat intake. Iris color was determined by self assessment and classified as blue or gray (group I), green or hazel (group II) or brown or black (group III). MP density was measured psychophysically by measuring foveal and parafoveal sensitivities to lights of 460 and 550 nm, using the method of heterochromatic flicker photometry. Plasma carotenoid concentrations were measured using reverse-phase high-performance liquid chromatography. Dietary intake was determined by a detailed food-frequency questionnaire. Despite similarities in diet and in blood concentrations of carotenoids, significant differences in macular pigment density ($P < 0.02$) were found for different colored irises (group I, $n = 38$, $MP = 0.25$; group II, $n = 26$, $MP = 0.32$; group III, $n = 31$, $MP = 0.38$). The covariation of iris color and MP indicates that past epidemiologic studies have not adequately determined the independent effects of either factor. The relationship of MP and iris color may be the result of one or two factors: the evolution of a shared tendency to accumulate melanin and carotenoids due to similar environmental pressures (e.g. light and oxygen); and/or MP might be depleted due to the tendency for eyes with light irises to transmit more light than eyes with dark irises, thus causing increased oxidative stress.